# Vasya Batmaev

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Ready to relocate

Long-form CV (for AI):
https://github.com/batmaev/batmaev/blob/main/CV/LLM.md

## Summary

ML engineer and fullstack developer from MIPT (leading Russian STEM university). I build algorithm-heavy products end-to-end: ML models (pytorch, sklearn), backends (FastAPI, Docker), and frontends (TypeScript, Svelte, Next).

## Skills

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|---|---|
| Daily | Python, TypeScript, git, pandas, LaTeX |
| Working knowledge | Julia, Wolfram Mathematica, MATLAB, C/C++, Swift, Kotlin |
| ML / Data Science | sklearn, pytorch, Bayesian inference, matplotlib, plotly |
| Infrastructure | FastAPI, Docker Compose, SQL, SQLAlchemy, Linux CLI |
| Languages | Russian (native), English (fluent) |

## Experience

**PhD Researcher – MIPT**
Challenge-response eye-movement biometric system (anti-deepfake). PyTorch/sklearn models on time-series data; FastAPI backend. Co-supervising a Master's student. Published 2025.

**CV Engineer – Nornickel (via Reksoft)**
IR video analysis for molten metal pouring: accretion and slag detection. Python, OpenCV.

**R&D Engineer – MIPT Telecom Lab**
Gyro-stabilized maritime antenna. IMU-to-quaternion pipeline with Kalman filters. Python hexapod control library, Plotly/Dash UI.

**Contract work:** Vependo VPN iOS port (KMP + Swift). Beverage kiosk UI in Electron + Svelte 5.

**Open source:** university email verification bot, 6,700+ users. Telegram API, SQLAlchemy.

## Education

**2024–2027: PhD,** Information Security – MIPT (Moscow Inst. of Physics & Tech).
Dissertation: Eye-movement biometrics on untrusted devices. Mostly online.

**2022–2024: M.Sc.,** Scientific Software – MIPT. GPA 5.0/5.0, honors.
Thesis: Bayesian analysis of variable-star data (Julia, Turing.jl).

**2018–2022: B.Sc.,** Applied Mathematics & Physics – MIPT. GPA 4.98/5.0, honors.
Thesis: Solving the Traveling Salesman Problem on a quantum computer.

- MIPT Deep Learning School
- Yandex Interface Development School: led frontend in a 24-person team (Next.js, OAuth)

## Awards

1st place in national student competition (yandex.ru/profi) in quantum technologies

## Publications

Defense against deepfake attacks via eye-movement liveness detection — Batmaev, Novikov // Information Security Issues, 2025

Dynamic eye-movement biometrics: trust with untrusted devices — Batmaev // MIPT Scientific Conference, 2025

Bayesian parameter estimation of symbiotic variable stars from light curves via probabilistic programming — Batmaev // MIPT Scientific Conference, 2024

Forest segmentation from satellite imagery via neural nets and mutual-information maximization — Matseyko, Batmaev et al. // IKI RAS Conference on Remote Sensing, 2023